

Aryan Avinash Kulkarni

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Skills

Languages: Java, Python, JavaScript, TypeScript, SQL

Technologies & Tools: Node.js, Express.js, FastAPI, Flask, PostgreSQL, MongoDB, React.js, Tailwind CSS, Docker, Git, GitHub, Vercel, Azure App Service

Machine Learning: Python, Pandas, NumPy, Scikit-learn, CatBoost, XGBoost, Feature Engineering, Time Series Forecasting, Model Evaluation

API Development: REST APIs, FastAPI, JSON Structured Data, LLM Workflows

Workflow & Deployment: Docker, Apache Airflow (Conceptual), Cloud Deployment, CI/CD Basics

Work Experience

Fundora, Mumbai Mar 2025 - Present

Backend Developer

- Designed and implemented scalable backend systems for startup and founder onboarding workflows using Node.js, Express.js, and PostgreSQL.
- Developed secure RESTful APIs for authentication, profile completion, verification flows, and seamless frontend-backend communication.
- Managed PostgreSQL schemas, optimized queries, and handled relational data design for startup, investor, and onboarding modules.
- Collaborated closely with frontend and UI/UX teams to deliver scalable production-ready backend architecture.

Dreamventz, Mumbai Jan 2025 - March 2025

Web Developer Intern

- Worked on enhancing and maintaining the existing event management platform using the MERN stack, improving website functionality and user experience.
- Developed and optimized the predefined event packages section, improving customer engagement and service discoverability.
- Assisted in Razorpay payment gateway integration and backend data handling for bookings, customer details, and platform operations.
- Performed website debugging, feature improvements, and frontend-backend integration for smooth platform performance.

Project Work

- **Trade Deficit Forecasting & Economic Intelligence Platform (2026)** Built a machine learning platform to forecast India's next-month trade deficit using monthly macroeconomic and trade data from 2011–2025. Engineered lag features, rolling averages, and macroeconomic interaction features for time-series forecasting. Compared Linear Regression, Random Forest, XGBoost, and CatBoost models, selecting CatBoost with best performance (RMSE $\approx$ 1.79B). Developed a full-stack dashboard using React and FastAPI with forecasting, scenario simulation, explainability, and live USD/INR + oil price integration.
- **Final Year Project – Evidence-Grounded Hallucination Detection for LLM Outputs (Ongoing)** Working on a system to detect hallucinated information generated by Large Language Models by extracting factual claims, retrieving supporting evidence, and assigning trust scores. Focused on trustworthy AI and evidence-grounded validation of LLM responses. Currently working on the Claim Extraction module using Selection, Disambiguation, and Decomposition inspired by Claimify.
- **Curely: Integrated Healthcare Platform** Developed a secure healthcare platform using the MERN stack, enabling patients and doctors to manage medical records, appointments, and healthcare workflows seamlessly.

Education

Vidyalankar Institute of Technology, Mumbai
Bachelor of Technology in Computer Engineering
Pursuing Honours in Next-Gen AI & Machine Learning

2023 - Present (2027)
CGPA: 9.92/10

Wamanrao Muranjan Junior College, Mumbai
Maharashtra State Board (HSC - 12th)

2021 - 2023
84.67%

Sou. Laxmibai English Medium School, Mumbai
Maharashtra State Board (SSC - 10th)

2014 - 2021
91.80%